A Point-Countable Norming-Family Obstruction for Norming M-Bases

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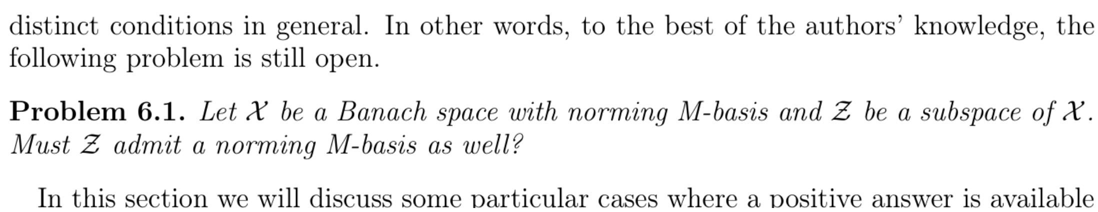
Source Problem

Hajek and Russo ask whether the existence of a norming Markushevich basis is hereditary to subspaces [1, Problem 6.1]. The problem appears on PDF page 24 of arXiv:2402.18442 and in the parsed source at `manuscript.tex:684-688`.



distinct conditions in general. In other words, to the best of the authors' knowledge, the following problem is still open.

Problem 6.1. *Let $\mathcal{X}$ be a Banach space with norming M-basis and $\mathcal{Z}$ be a subspace of $\mathcal{X}$. Must $\mathcal{Z}$ admit a norming M-basis as well?*

In this section we will discuss some particular cases where a positive answer is available

Result

This packet records a sharp obstruction to a natural reduction route for Problem 6.1. The result does not solve the heredity problem. It shows that the restricted-coordinate data inherited by a subspace is not, by itself, enough to force a norming M-basis.

Theorem 1. *There is a Banach space Y and a family $(g_\alpha)_{\alpha<\omega_1} \subseteq Y^*$ such that:*

1. $\operatorname{span}\{g_\alpha : \alpha < \omega_1\}$ *is* 1*-norming for* Y*;*
2. *for every* $y \in Y$*, the set*
$$\{\alpha < \omega_1 : g_\alpha(y) \neq 0\}$$
is countable, and hence $(g_\alpha(y))_{\alpha<\omega_1} \in c_0(\omega_1)$*;*
3. Y *has no norming M-basis.*

Proof Intuition

The point evaluations on $C_0([0,\omega_1])$ are as close as possible to the coordinate family one hopes to obtain by restricting a norming M-basis: finite linear combinations of them norm the sup norm, and each vector has only countably many nonzero coordinates. The reason this still does not yield a norming M-basis is that finite-span norming coordinates are weaker than biorthogonalizable norming coordinates. Alexandrov–Plichko's theorem supplies the obstruction: $C([0,\omega_1])$, and hence its isomorphic hyperplane $C_0([0,\omega_1])$, admits no norming M-basis.

Proof

Let

$$Y = C_0([0, \omega_1]) = \{f \in C([0, \omega_1]) : f(\omega_1) = 0\}$$

with the sup norm. For each $\alpha < \omega_1$, put

$$g_\alpha = \delta_\alpha|_Y.$$

First, $\operatorname{span}\{g_\alpha : \alpha < \omega_1\}$ is 1-norming for Y. Indeed, for $f \in Y$,

$$\|f\|_\infty = \sup_{\alpha \le \omega_1} |f(\alpha)| = \sup_{\alpha < \omega_1} |f(\alpha)| = \sup_{\alpha < \omega_1} |g_\alpha(f)|.$$

Since each g_α has norm at most 1, the finite linear span of the g_α's is 1-norming.

Second, every $f \in Y$ has countable support with respect to this family. By continuity at ω_1 and the fact that $f(\omega_1) = 0$, for every $n \ge 1$ there is an ordinal $\beta_n < \omega_1$ such that

$$|f(\alpha)| < 1/n \qquad (\beta_n \le \alpha \le \omega_1).$$

Let $\beta = \sup_n \beta_n$. Since ω_1 has uncountable cofinality, $\beta < \omega_1$. Hence $|f(\alpha)| < 1/n$ for every n whenever $\alpha \ge \beta$, so $f(\alpha) = 0$ for all $\alpha \ge \beta$. The support of f is therefore contained in the countable ordinal segment $[0, \beta)$. In particular $(g_\alpha(f))_{\alpha<\omega_1} \in c_0(\omega_1)$.

Finally, Y has no norming M-basis. Hajek and Russo recall the theorem of Alexandrov–Plichko that $C([0, \omega_1])$ admits no norming M-basis [1, Theorem 4.2]. The same source also notes that $C_0([0, \omega_1])$ is isomorphic to $C([0, \omega_1])$ [1, Example 4.1]. Existence of a norming M-basis is invariant under Banach-space isomorphism: pulling a norming M-basis through an isomorphism preserves biorthogonality, fundamentality, totality, and norming up to the equivalence constant of the isomorphism. Therefore Y admits no norming M-basis.

Consequences for Problem 6.1

For a subspace Z of a Banach space with a norming M-basis, the restricted coordinate functionals do form a norming c_0-valued family. The theorem above shows that this abstract fact alone cannot prove heredity. Any full proof of Problem 6.1 must use additional structure from the ambient biorthogonal system, such as a bounded reconstruction, projectional, or finite-span biorthogonalisation mechanism. Conversely, this is not a counterexample to Problem 6.1, because the known Russo–Somaglia extension of Alexandrov–Plichko implies that $C([0, \omega_1])$ does not embed into a Banach space with a norming M-basis [1, Section 6].

Verification Notes

The proof uses only standard facts recorded in the source survey: Alexandrov–Plichko's theorem for $C([0, \omega_1])$, and the isomorphism between $C([0, \omega_1])$ and $C_0([0, \omega_1])$. The countable-support calculation is included above.

Limitations and Novelty Check

This packet does not answer the original heredity problem and should not be counted as a counterexample to it. It disproves only the broader abstract principle:

$$\text{norming point-countable coordinate family} \Longrightarrow \text{norming M-basis.}$$

The run indexes were checked for arXiv:2402.18442 and the existing local partials before promotion. The obstruction is based on the source paper's own discussion of $C([0,\omega_1])$, but the use of point evaluations on $C_0([0,\omega_1])$ is recorded here as a route-blocking lemma for this run.

Human Review Recommendation

Review as a route obstruction. In particular, check that the theorem is not accidentally read as saying $C_0([0,\omega_1])$ embeds in a norming-M-basis space; it does not. Its role is to show that the inherited restricted coordinate data from an arbitrary subspace would need additional ambient structure before it can be biorthogonalized into a norming M-basis.

References

[1] P. Hajek and T. Russo, *Norming Markushevich bases: recent results and open problems*, arXiv:2402.18442.

[2] A.G. Alexandrov and A.N. Plichko, *Connection between strong and norming Markushevich bases in nonseparable Banach spaces*, Mathematika **53** (2006), 321–328.